VENTILATION NETWORK ANALYSIS

A mine ventilation network is a mathematical representation of a real-world mine ventilation system.

- It is an assembly of pathways for mine air and the data associated with it, e.g. resistance, flow rate etc.

- These pathways, also called airways or branches, are represented by lines and are interconnected among themselves at the points called junctions or nodes.

Fundamentals of Fluid Networks

The basic terms used in ventilation network are

1. Terminologies:

a. A Ramified Network: is a closed and interconnected system of branches through which the fluid may flow.

In a mine ventilation network, the fluid is air and surface-connecting branches such as shafts are effectively closed through the free atmosphere.

b. A junction is a point at which three or more airways meet.

However, the ventilation engineer may define any specified point in the network as a junction.

c. A Branch is a single airway connecting two junctions. It follows that any one branch have junctions only at its end points.

(d) A mesh (or loop) is any closed path of connected branches within the network.

Therefore, a ventilation network can be divided into branches, junctions and meshes which are defined as foll

A 'branch' is an airway between two junctions
A 'junction' is a point where three or more airways meet
A 'mesh' is a closed path traversed through a network of airways.

Techniques for the solution of ventilation networks.

~~Four methods may be employed for solving ventilation networks~~

The calculation of the resistance or pressure loss in a split easy if it ~~is connected~~ consists of branch airways so connected that the whole network can be resolved into a simple series-parallel circuit. In practice, however,

ventilation networks are usually complex and need special methods for solution.

Four methods may be employed for solving ventilation netwo

1. The first of these involves the compounding of series- and parallel connected airways into single equivalent resistances.

2. The second method is the analytical solution of of equations obtained by direct application of Kirchhoff's laws.

3. The third is the use of physical models or analogue to represent the mine ventilation system and

4. Finally, the methods of successive approximation have come to the forefront with the evolution of electronic digital computers.

1. Equivalent Resistance Technique.

This technique is useful for those networks, or parts of networks, which are made up of a number of airways connected in series and/or in parallel. Such systems may be defined as "simple" networks. Conversely, systems which can not be resolved into a series/parallel configuration may be defined as "complex" networks. Unfortunately, very few mines can be described completely in terms of series/parallel connections. The equivalent technique is, nevertheless, very useful for airflow investigations in localised parts of a mine.

2. Direct Analysis using Kirchhoff's laws

This analysis involves the use of Kirchhoff's laws. Kirchhoff's electrical laws (laws for electrical circuits) holds good for airflow circuit and hence applicable for solving complex network problems.

Kirchhoff's First Law:

It states that the algebraic sum of all mass flow rates at any junction is zero, that is,

$$\sum_{i=1}^{n} M_i = 0 \quad (1)$$

where M_i = mass flow rate, kg/s

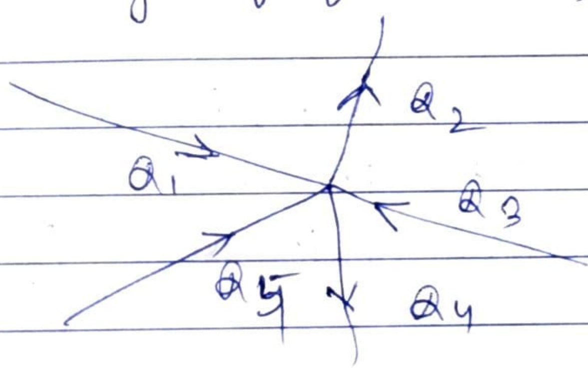
$= \rho_i \cdot Q_i$ kg/s

where ρ_i and Q_i are the air density and volume flow rate, respectively in the ith branch connected to the junction, there being n such branches.

Where changes in density are small or all measurements have been corrected to a standard density, eqn. 1 reduces to

$$\boxed{\sum_{i=1}^{n} Q_i = 0} \quad \text{—— (2)}$$

Therefore, Kirchhoff's 1st law states that the algebraic sum of the quantity flowing into a junction is equal to zero, or in other words the total quantity of air flowing into a junction of airways equals that flowing out of it.



or.

$Q_1 + Q_3 + Q_5$
$= Q_2 + Q_4$

Therefore, the sum of all air quantities flowing towards a junction must equal the sum of all air quantities flowing away from that junction.

Kirchhoff's Second Law:

MEMO / TO DO It states that the algebraic sum of all frictional pressure drops around any ~~etc.~~ closed mesh is zero.

If any fan and/or natural ventilation pressure is present, the algebraic sum of of all frictional pressure drops around any closed mesh, less any fan and natural ventilating pressure, is equal to zero.

i.e

In case of a mesh containing a fan and natural ventilating pressure,

$$\boxed{\sum_{i=1}^{m} (P_i - P_{fi}) - (NVP)_m = 0} \longrightarrow (3)$$

In the simplest case of a mesh containing no fan and no natural ventilating effects, Kirchhoff's second law reduces to

$$\boxed{\sum_{i=1}^{m} P_i = 0} \longrightarrow (4)$$

where

P_i = frictional pressure drop, Pa in i-th ~~branch~~ mesh

P_{fi} = Difference in total pressure across the fan, Pa

$(NVP)_m$ = Natural ventilating pressure, Pa in m meshes.

Direct Analysis

This technique involves writing down

- Kirchhoff's first law (eqn. (2)) for each junction in the network and

- Kirchoff's second law (eqn. 3) for each mesh.

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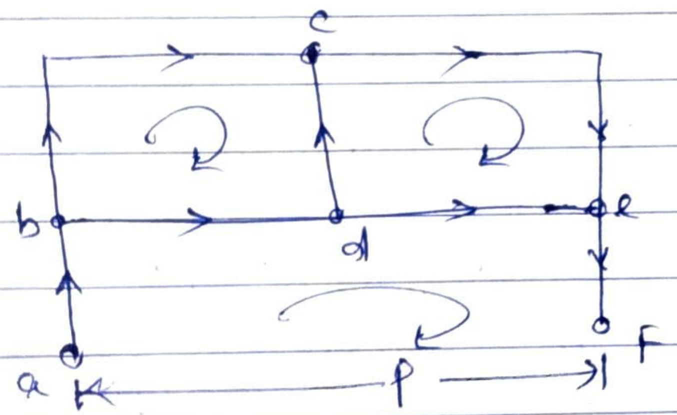
MEMO / TO DO

If a network contains 'b' branches, 'j' junctions, there will be b equations ~~and~~ to determine and therefore, b equations have to be solved, to ~~solve the~~ determine all the volume flow rates passing through ~~each b bran~~ 'b' branches.

– From the first law, ~~of~~ j' equations can be written. However, since the volume flow rates at the jth junction are already defined by the volume flow rates at other junctions, only (j-1) of the j' equations are independent.

Therefore, b − (j−1) equations remain to be written and are obtained by choosing a minimum of (b−j+1) meshes and using the second law for each of them.



Let us consider a simple network as shown above. A fan generating a pressure p is situated across a and f.

Now from Kirchhoff's first law,

considering junction 'b'.

$Q_{ab} = Q_{bc} + Q_{bd}$ or $Q_{bd} = Q_{ab} - Q_{bc}$

junction d : $Q_{bd} = Q_{dc} + Q_{de}$ or $Q_{de} = Q_{bd} - Q_{dc}$

MEMO / TO DO

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junction c : $Q_{ce} = Q_{bc} + Q_{dc} \Rightarrow Q_{dc} = Q_{ce} - Q_{bc}$

junction e : $Q_{ef} = Q_{ce} + Q_{de} \Rightarrow Q_{de} = Q_{ef} - Q_{ce}$

$= Q_{ab} - Q_{ce}$ since $Q_{ab} = Q_{ef}$

Applying Kirchhoff's second law in mesh abdefa

$$R_{ab} \cdot Q_{ab}^2 + R_{bd} \cdot Q_{bd}^2 + R_{de} \cdot Q_{de}^2 + R_{ef} \cdot Q_{ef}^2 - P = 0$$

$$\Rightarrow (R_{ef} + R_{ab}) \cdot Q_{ab}^2 + R_{bd}(Q_{ab} - Q_{bc}) + R_{de}(Q_{ab} - Q_{ce})^2 - P = 0$$

in Mesh bcdb, $R_{bc} Q_{bc}^2 - R_{dc} \cdot (Q_{ce} - Q_{bc})^2 - R_{bd} \cdot (Q_{ab} - Q_{bc})^2 = 0$

In Mesh cedc, $R_{ce} Q_{ce}^2 - R_{de}(Q_{ab} - Q_{ce})^2 + R_{dc}(Q_{ce} - Q_{bc})^2 = 0$

where R = Resistance of the branch and the suffixes indicate the branch concerned.

Q = Quantity flowing in any branch.

From the above three equations, it is possible to solve the three unknowns, Q_{ab}, Q_{bc} and Q_{ce}. For ~~equations~~ networks with larger no. of branches, a large no. of equations have to be solved.

In the above example,

no. of branches, b = 8 (including the fan as a branch)

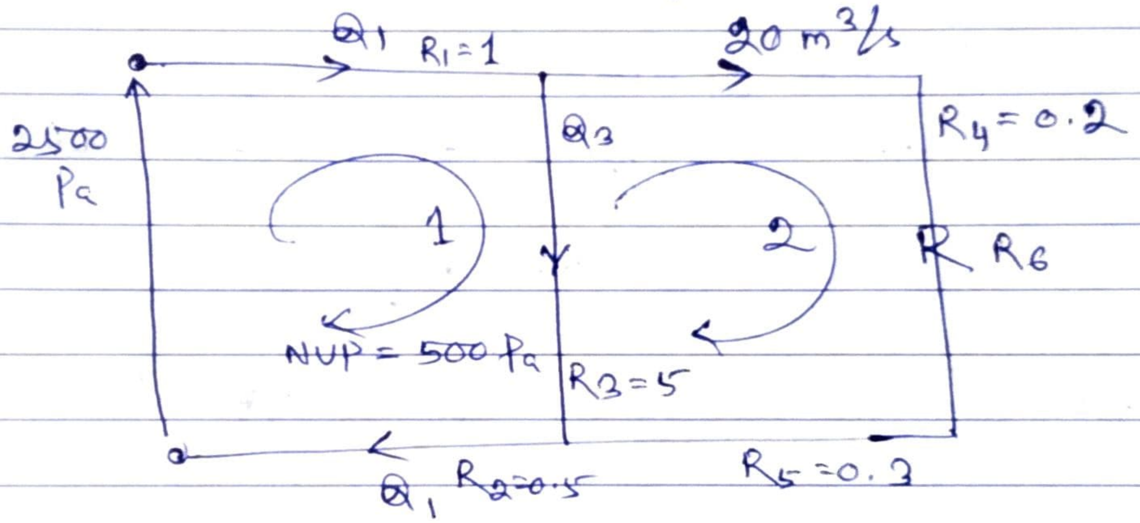
no. of junctions, j = 6 (each letter in fig. indicates a junct'n)

Therefore,

No. of meshes, $m = b - j + 1 = 8 - 6 + 1 = 3$.

Example.

A two-mesh network is shown diagrammatically in the following figure. A differential pressure of 2500 Pa is applied across the circuit and a natural ventilating pressure of 500 Pa acts in the direction of airflow within mesh 1. A regulator, R_6 is constructed in the right most branch in order to limit the airflow in that branch to 20 m^3/s. Given the resistance of all airways, find the distribution of airflow and the resistance of the regulator.



Soln. The total quantity of airflow entering the circuit is Q_1 (say)

The flow around the regulated branch = 20 m^3/s

Then $Q_3 = Q_1 - 20\ m^3/s$ is the flow through the centre branch

Thus Kirchhoff's first law is satisfied.

The choice of meshes is quite arbitrary provided that the minimum no. $(b-j+1)$ meshes is satisfied and that all branches are included in the mesh pattern.

A third mesh $R_1 - R_4 - R_6 - R_5 - R_2$, may equally well have been choosen.

Similarly, the choice of positive direction around each mesh is arbitrary, provided that the direction chosen for each mesh is adhered to throughout the analysis.

Applying Kirchhoff's second law to each of the chosen meshes and assuming the square law ($P = RQ^2$) to hold the following equation can be constructed:

Mesh 1 : $R_1 Q_1^2 + R_3 (Q_1 - 20)^2 + R_2 Q_1^2 - P_f - NVP = 0$

or $1 Q_1^2 + 5 (Q_1 - 20)^2 + 0.5 Q_1^2 - 2500 - 500 = 0$

$\Rightarrow 6.5 Q_1^2 - 200 Q_1 - 1000 = 0$

This is a straightforward quadratic equation which can be solved to give.

$$Q_1 = 35.147 \ m^3/s$$

Therefore,

the flow through the R_3 branch = $15.147 \ m^3/s$

Mesh - 2 : $R_4 20^2 + R_6 20^2 + R_5 20^2 - R_3 (Q_1 - 20)^2 = 0$

(Note the change of sign for the R_3 branch, due to the ~~change~~ mesh direction being opposite to the branch flow)

$\Rightarrow (0.2 + R_6 + 0.3) 400 \cancel{+ 400} - 5 (15.147)^2 = 0$

$\Rightarrow R_6 = 2.368 \ Ns^2/m^8$

If the specified flow of $20 \ m^3/s$ necessitates the use of a booster fan instead of a regulator in the right-hand branch, then a similar procedure may be used to determine the reqd. booster fan pressure.